

WELL IDENTIFICATION & LOCATION METRICS

Well Name:	Duliajan-99 (DUL-99)	Field:	Duliajan Field
Coordinates:	26.848° N, 95.332° E	Loss Depth:	3,210 m (MD)
Formation Target:	Sylhet Karst Limestone	Pre-Loss Mud Weight:	1.32 g/cc
Hole Size:	8.5-inch section	Bit Type:	PDC 5-Blade Matrix

TOTAL MUD LOSS INCIDENT IN FRACTURED LIMESTONE

At 3,210m depth, bit penetrated a natural vuggy fracture network within the Sylhet Limestone formation. Mud flow-rate return at shaker dropped to ZERO (100% total lost circulation). Fluid level in the annulus dropped rapidly by 120m, threatening well control by losing hydrostatic column.

■ **HAZARD CLASSIFICATION:** Severe Lost Circulation in Karstified / Fractured Sylhet Limestone.

MITIGATION & RECOVERY EXECUTED

- Annular Top-up:** Immediately filled annulus from top with fresh water to maintain hydrostatic pressure.
- Coarse LCM Pill:** Mixed and pumped 60 bbl Loss Circulation Material (LCM) pill containing Nut Plug (Coarse), Mica (Medium), and Calcium Carbonate (25 lbg/bbl).
- Cement Squeeze Plug:** Pumping 35 bbl Class-G thixotropic cement plug across 3,200m–3,210m interval. Allowed cement to set for 12 hours.
- Drilling Resumed:** Drilled out cement plug; full returns restored with 1.22 g/cc reduced mud weight.

TOTAL NPT ELAPSED: 44.0 Hours | **LCM & CEMENT MATERIAL COST:** ■18.5 Lakhs